InventOR: Can Large Language Models Set Reorder Points and Safety Stocks? Industrial Evidence Against ERP and Data-Driven Benchmarks

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Abstract

Should the reorder point, safety stock and order quantity of a plant-material pair be revised, and can a large language model (LLM) do it? Parameters often remain at ERP values set long ago because revising them takes scarce analyst time. We evaluate an LLM configured only through a prompt, without training, inside a reviewed workflow called *InventOR*: the LLM proposes a reorder-point/order-quantity (r, Q) policy with a written justification, deterministic software gates and replays it on held-out demand, and a planner decides. Using 365 plant-material pairs from three plants of an industrial manufacturer, three LLM runs, seven reference policies (including textbook code that applies the formulas the prompt prescribes) and an ablation withholding the ERP planning parameters, we compare cost, service and inventory in a hard-limit storage replay. The LLM costs about 11% less than ERP-derived parameters, but 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile baselines. It reproduces the textbook formulas (median safety-stock and order-quantity ratios of 1.00), with a 9% higher reorder point and about 10% run-to-run variation, whereas code given the same ERP inputs costs 44% more. Once lost sales are charged at 1% of unit cost or more, most policies lie within 4% of each other and only a simulation-tuned policy is clearly cheaper. Simulated trajectories exceed storage limits for

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53–66% of pairs that pass static checks. The LLM is a feasible, reviewable generator of parameters, not an optimizer; planner benefits remain untested.
Keywords: Inventory Control, Large Language Models, Generative Artificial Intelligence, Safety Stock, Intermittent Demand, Empirical Evaluation

1. Introduction

Replenishment parameters in manufacturing are set per plant-material pair: a reorder point, a safety stock and an order quantity. In large ERP systems these values often remain at levels set years earlier, while demand, supplier performance and storage constraints change. Revisiting them requires analysts who can fit demand and lead-time statistics, apply inventory theory, respect minimum order quantities (MOQs) and storage limits, and explain the result to planners. A conversational large language model (LLM) with a code interpreter appears to offer this scarce skill, and a growing literature explores such tools for supply-chain and operations decisions [1, 2, 3, 4, 5, 6], building on the long-standing promise of data analytics in logistics [7].

Three problems limit its use. First, LLM outputs can be unfaithful or fabricated [8], so a proposed policy is a candidate, not a decision. Second, superiority claims over ERP parameters are hard to interpret when the LLM and the comparators receive different information. Third, a storage rule that holds when a policy is written down can be violated as stock builds up over time.

Prior LLM work in operations mostly translates problem statements into solver code or studies agents on synthetic benchmarks (Section 3); we found no evaluation in which an LLM sets per-material (r, Q) parameters for an industrial portfolio and is compared with ERP values and with strong data-driven baselines under controlled information. This paper provides one. We call the workflow *InventOR*: for each plant-material pair, an enterprise

LLM application that is configured only through a prompt and receives no project-specific training or fine-tuning proposes a candidate (r, Q) policy together with a justification and an escalation flag. Deterministic software parses, gates and simulates every proposal, and a planner reviews it and alone decides whether any ERP parameter is changed (Section 2).

We study five questions on real industrial data: what cost, service and inventory the LLM policies achieve relative to ERP-derived parameters, to data-driven baselines and to textbook code that applies the formulas the prompt prescribes, when the information available to each approach is controlled (RQ1); why the LLM differs from that code, in terms of formula fidelity, run-to-run noise, demand segments and the price of shortages (RQ2); whether the policies respect storage limits along simulated trajectories (RQ3); how stable they are across independent runs (RQ4); and in which plants and demand regimes a planner should trust, escalate or reject a proposal (RQ5).

We contribute (C1) a workflow with explicit requirements that bound the role of the LLM (Sections 2 and 4); (C2) an information-controlled, trajectory-level evaluation with seven reference arms, including deterministic code that applies the prompt's own formulas, an ablation denied the ERP planning parameters and a hard-limit storage replay (Section 5); (C3) complete-cohort evidence from three LLM runs on 365 plant-material pairs, including the cost decomposition, a break-even analysis of the shortage price and a mechanism analysis of why the LLM does not beat code (Section 6); and (C4) implications for practice and research perspectives (Section 7).

Our main findings are as follows. Against ERP-derived parameters the LLM policies are about 11% cheaper at a mean fill rate 0.1–0.4 points lower, but the LLM costs 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile (r, Q) baselines at essentially equal service, and its saving over ERP comes from carrying less inventory at the

price of more stockout days. Storage limits that hold in a static check are violated along unconstrained trajectories for 53–66% of pairs. The LLM reproduces the prompt's textbook formulas almost exactly (median safety-stock and order-quantity ratios to the code arm of 1.00, reorder points 9% higher), so its differences from code are small and noisy, while code that receives the same ERP inputs as the LLM costs 44–45% more than the LLM. When lost sales are charged at 1% of unit cost or more, the ERP-derived, history-only, Croston/SBA, empirical-quantile and LLM policies lie within 4% of each other and only a simulation-tuned policy is clearly cheaper. The simulation therefore does not show superior optimization by the LLM. If only parameters are needed, running the formulas as code is cheaper and exactly repeatable; the LLM adds a written justification, an escalation flag and no coding effort, and whether that helps planners remains to be tested.

2. Problem Setting and Workflow

2.1. The decision problem

The decision supported is the periodic (re)setting of a reorder point, a safety stock and an order quantity for each plant-material pair held in an ERP system. It is a partly structured operational-control decision [9]: the calculation is textbook, but the choice of estimator, the treatment of intermittent demand, the plausibility of recorded lead times and the handling of constraints require judgement that is applied inconsistently when thousands of materials compete for a few analysts' time. The decision-maker is the material planner; hard constraints are the minimum order quantity (MOQ) and a physical storage limit; and feedback is slow, because a changed parameter shows its effect only in later demand. The scope is one pair at a time (single item, single echelon, continuous review, lost sales, (r, Q) family); order release, supplier selection, forecasting-model development and multi-echelon networks

are out of scope, and the LLM is a candidate generator that does not forecast, place orders, simulate or decide.

2.2. The InventOR workflow and its level of automation

> **Definition.** *InventOR is a human-reviewed planning workflow that, for one plant-material pair at a time, (i) obtains from a large language model, configured only by a written prompt, a candidate (r, Q) policy with a justification and an escalation flag; (ii) verifies the candidate deterministically against explicit constraints; (iii) evaluates it by counterfactual simulation against reference policies on held-out demand; and (iv) presents the result to a planner, who alone decides whether ERP parameters are changed.*

In the levels of automation of Parasuraman et al. [10], InventOR operates at a low-to-intermediate level: it suggests one alternative and leaves execution to the human. Accepting suggestions invites complacency and automation bias [11], which the review rules of Section 4.3 are meant to counter.

2.3. Requirements on the workflow

The requirements follow from the known unreliability of LLM outputs [8], guidance on human–AI interaction and accountability [12, 13], run-to-run nondeterminism of LLM inference [14] and the constraints of inventory control. Table 1 ties each to a design feature (Section 4) and an evaluation (Section 6).

Table 1: Requirements on the workflow, the features that address them and where they are evaluated.

ID	Requirement	Design feature	Evaluation
DR1	Bounded generation: the LLM proposes and has no authority	Narrow candidate-generator role; prohibited methods; no order placement	Sections 6.1, 6.3
DR2	Verifiable feasibility along time, not only at a point	Deterministic gate; hard-limit trajectory replay	Sections 6.1, 6.5
DR3	Counterfactual evidence for the reviewer	Held-out simulation against reference arms	Sections 6.2, 6.10
DR4	Information transparency: comparisons control what each arm sees	Seven reference arms; LLM ablation without ERP parameters	Sections 6.3, 6.4
DR5	Reproducibility and accountability of the configuration	Frozen prompt file; per-call log; repeated runs	Section 6.8
DR6	Human control with explicit escalation	Trust, escalate and reject rules; escalation flag	Sections 4.3, 6.6

3. Related Work and Positioning

Human–AI decision making in planning.. Decision support aims to improve the judgement of a human decision-maker, not to replace it [9, 15, 16], and how suggestions are scoped, presented and verified determines whether users rely on them appropriately [10, 11]. In inventory control, human override of algorithmic safety-stock advice is a documented risk [17], people discount algorithms after seeing them err [18], and AI-based route planning yields efficiency advantages when it is followed in full [19]. InventOR therefore keeps the planner as decision-maker and makes the evidence for each proposal explicit.

LLMs in operations and supply-chain decisions.. Reviews and frameworks map the opportunities of generative AI in supply chains [1, 2, 3, 20, 4, 5] and stress human roles, data quality and governance [6, 21, 22]. Evidence from manufacturing planning is thin: a fine-tuned LLM for digital-manufacturing optimization [23], a transformer trained to set order timing and quantity [24] and evidence that ChatGPT reproduces managers' decision biases [25]. Most technical LLM work translates problem statements into solver code [26, 27, 28, 29] or studies multi-agent control in inventory flows [30, 31]. The closest conceptual neighbour studies how OR algorithms, LLM agents and humans complement each other in multi-period inventory control on a benchmark of more than 1,000 instances, and reports that OR-augmented LLM

methods outperform either alone [32]; we ask the prior question of how far an LLM alone gets on industrial data. In forecasting, LLMs do not consistently outperform human judgment in retail [33], and an LLM forecasting copilot for spare-parts demand shows high variability for irregular demand that averaging repeated queries reduces [34]. We test run-to-run variability directly (Section 6.8) and let the LLM propose a complete policy, not a forecast.

Data-driven inventory control without LLMs.. Learning-based replenishment includes policies learned from non-stationary demand [35], forecasts with leading indicators [36], deep reinforcement learning for omnichannel and multi-echelon systems [37, 38, 39], contextual robust optimization [40] and reinforcement learning combined with demand forecasting to set order points [41], while heuristics remain competitive in constrained settings [42]. The closest operational neighbour is a trained framework that jointly sets replenishment parameters, safety stock and safety lead time on real automotive data [43]. Forecasting research links forecasts to inventory parameters through safety stock under causal forecasts [44], combined-forecast error distributions [45] and forecasting models optimized for inventory cost [46]. In *Expert Systems with Applications*, model-based deep reinforcement learning has been used for new products with short histories [47], offline-trained agents adapt online to unseen supply-chain configurations [48], and further papers propose a temporal-fusion-transformer replenishment strategy [49], graph recurrent networks for store-SKU replenishment [50] and time-series mixer models for demand forecasting and inventory [51]. Recent articles there also combine LLMs or agents with supply-chain decision support, as a hybrid fuzzy-LLM system for lifecycle-oriented strategy [52] and a governed multi-agent framework for resilient supply-chain decisions [53]. In a title search of the journal we found no evaluation in which a prompt-configured LLM sets per-material (r, Q) parameters against ERP values and textbook code. These methods need training or tuning; InventOR needs none, but its generator

cannot be inspected beyond its written justification.

Intermittent and spare-parts demand.. Manufacturing histories are often intermittent or lumpy. Work began with the bias of Croston's estimator and its stock-control performance [54, 55] and continues with neural [56], bootstrapping [57], obsolescence-aware [58], temporal-hierarchy [59] and state-space [60] forecasts, order-up-to control [61], safety-stock placement [62] and lead-time forecasting [63], all of which bear on the inputs of the policies we compare. The wider literature adds specialized estimators and classification schemes [64, 65, 66, 67, 68], spare-parts forecasting [69] and multi-echelon spare-parts models [70]. InventOR is not a new inventory model and does not claim to beat trained or optimized methods; its (r, Q) policy is the standard workhorse of continuous-review control [71, 72, 73].

4. The InventOR Workflow

4.1. Architecture and components

Following the classical decision-support components [15, 16], InventOR has a *data* component (a material-scoped extract of daily demand, lead times, working-day flags, cost and constraint parameters and ERP reference values; the fields supplied are controlled and logged; DR4), a *knowledge* component (the system prompt file defining the task, the prohibited methods, the storage rules and the output schema, Supplementary File S1; DR1, DR5), a *model* component split into a generator (an enterprise LLM application with a code interpreter, configured only by the prompt, proposing one (r, Q) policy with a justification and an escalation flag; DR1) and a deterministic verifier and simulator (parser, gate, held-out simulation with hard storage-limit replay and reference arms; DR2, DR3), a *dialog* component (policy, justification, flag and simulated comparison shown to the planner under the trust, escalate and reject rules; DR6) and a *governance* layer (frozen prompt file, per-call log

of input hash, time, session and raw response, repeated runs; DR5). The generator is the only non-inspectable component. In the evaluated prototype the dialog component is the per-pair output record; no interactive interface was built or evaluated.

4.2. Operation in four steps

For one material, InventOR answers how much safety stock to hold, at what level to reorder and how much to order, in four steps (Figure 1).

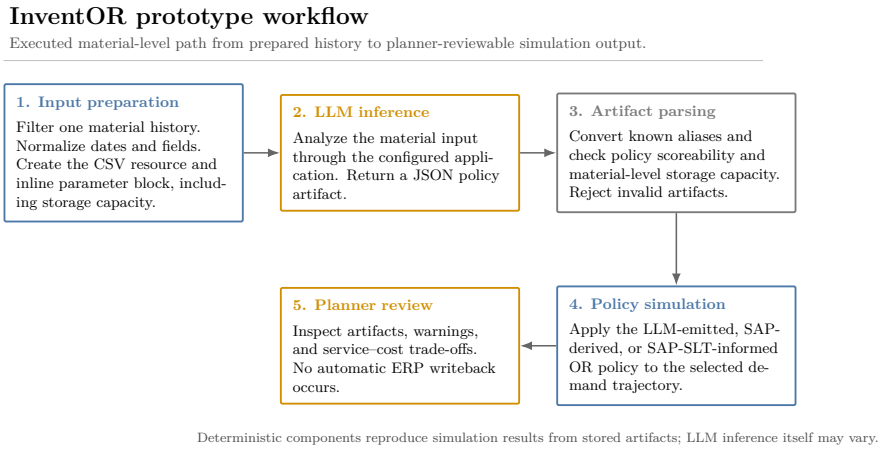


Figure 1: InventOR workflow: material data are prepared, an LLM proposes a policy, deterministic software parses, checks and simulates it, and a planner reviews the result.

Step 1. Generate. The LLM, accessed through an enterprise application with a code interpreter, receives one material's history (daily demand, realized lead times, working-day flags) and a short parameter block (unit cost, holding rate, ordering cost, MOQ, service target, storage limit, and current ERP safety stock and safety lead time for comparison). A written prompt (the only configuration; no weights are changed) [74] instructs it to compute demand and lead-time statistics, apply an economic-order-quantity (EOQ) argument and a normal-approximation safety stock, respect the storage limit, and report a structured policy with a short justification and escalation flag.

Step 2. Gate. Deterministic code extracts the policy, rejects malformed or infeasible outputs (missing or non-numeric fields, non-positive reorder point or order quantity, negative safety stock, reorder point below safety stock, safety stock plus order quantity above the storage limit) and retries the generation up to three times. The backtest applies one further admission check, that the order quantity is not below the MOQ.

Step 3. Simulate. Each accepted policy is replayed on the material's demand observed *after* the data cutoff, a held-out evaluation in the sense of out-of-sample forecast testing [75, 76], together with the comparator policies, producing holding and ordering cost, fill rate and stockout days.

Step 4. Review. A planner sees the policy, the LLM's justification and flag, and the simulated comparison. Only the planner changes ERP parameters. The planner's decision (accept, adjust or reject), the review time and an optional free-text comment are logged with the call, so that practitioner feedback accumulates as a by-product of use; the prototype evaluated here does not yet implement this logging.

The LLM is a *candidate generator* whose output is never trusted without the gate; it does not forecast, place orders or run the simulation. The prompt prohibits Monte Carlo simulation, distribution fitting, machine-learning libraries and specialized intermittent-demand estimators, so that the policy rests on transparent statistics.

4.3. Trust, review and reject rules

We propose three rules for planners, in line with guidelines for human–AI interaction and AI auditing [12, 13]. The rules and the 0.5-point threshold are design proposals not

evaluated with planners; the trust rule would be met by a minority of pairs (the LLM is cheaper than the history-only OR in 74–77 of 346).

Trust (apply after review) when the proposal passes the gate, the simulated trajectory respects the storage limit when orders are truncated at it, and the simulated cost is not above that of the best history-only OR reference (Croston/SBA or empirical quantile) with a fill rate not more than about 0.5 points lower.

Escalate when the LLM raises an escalation flag, the material is lumpy or has strongly seasonal or trending demand, the plant or calendar data were repaired, or the policy differs strongly between repeated generations.

Reject when the proposal fails the gate after the retry budget, exceeds the storage limit in the hard-limit replay, or is dominated (higher cost and lower fill) by a history-only reference.

4.4. Provenance and reproducibility of the configuration

The system prompt, a plain-text file deployed unchanged in the enterprise application, defines the task, forbidden methods, capacity rules and output schema (Supplementary File S1). Every call logs the SHA-256 of its input, UTC time, session identifier and raw response. The platform does not expose the model identifier or sampling settings; the evaluated model (Anthropic Claude Sonnet 4.6 with a code interpreter) is operator-declared. Reproducibility rests on the prompt file, the call logs and three independent runs (Section 6.8).

5. Application and Evaluation Design

5.1. Evaluation approach

The evaluation addresses the requirements of Table 1. The primary basis is the hard-limit replay with working-day receipts and no shortage penalty; sensitivity analyses vary

these choices. The evaluation is an ex post simulation study: real data and a faithful simulation but no live users. The prompt prescribes a closed-form recipe (an economic order quantity with a minimum-order-quantity floor and a normal-approximation safety stock), so the history-only, SLT-informed and ERP-floor OR arms of Section 6.3 are the deterministic-code counterparts of what the LLM is asked to do, with the same formulas and, depending on the arm, the same inputs. It establishes technical efficacy, feasibility and reliability, not usability, adoption or improved planner decisions.

5.2. Industrial application and data

The data come from a large manufacturer, and the evaluation is retrospective. The dataset contains 233,078 daily records for 411 plant-material pairs from three plants (A, B and C), with a data cutoff on 10 March 2026 and a post-cutoff horizon observed to 27 August 2026. Of the 411 pairs, 365 are eligible for scoring (234 in Plant A, 65 in Plant B, 66 in Plant C) and are sent to the LLM. For 19 pairs (all in Plant A) the MOQ plus the reference safety stock exceeds the storage limit, so no comparator can satisfy both rules; these are excluded from outcome comparison, leaving a *common-feasible cohort* of 346 pairs (215, 65 and 66). Demand is concentrated (the five largest materials account for 40.3% of demand). The source working-day flag of Plant C is all zero; we reconstruct it from the same-date majority of Plants A and B and test this repair in Section 6.10. Data quality is a known threat to decision support [77] and inventory records [78, 79, 80].

Each pair is sent as a material-scoped CSV with daily demand, recorded lead times, a working-day flag, the minimum order quantity, unit cost, holding-cost rate, ordering cost, service-level target, the storage limit, and two *ERP planning parameters* (current safety stock and safety lead time), which are withheld in the ablation arm. The CSV also carries supplier and lane context fields that the prompt does not use. A masked version of the

panel is released for replication (Section 6.9); there the plants are labelled PL1, PL2 and PL3 (corresponding to A, B and C).

5.3. Reference arms and information content

ERP safety stock and safety lead time embed planner judgement and are alternative ways of buffering unreliability [81], and planned lead times can drift upward [82]. The LLM and the classical comparators therefore do not necessarily use the same information, and we construct arms that make it explicit (Table 2).

Table 2: Comparison arms and the information each uses. All arms respect MOQ and are projected to the storage limit when possible.

Arm	Parameters from	ERP inputs used	Purpose
SAP-derived	ERP safety stock; EOQ order quantity	Safety stock	ERP-derived reference (not observed practice)
SLT-informed OR	Stationary (r, Q) with ERP safety lead time added to the lead time	Safety lead time	OR policy using the ERP safety lead time
ERP-floor OR	SLT-informed OR with safety stock floored at the ERP value	Both	OR arm matched to the LLM's full information
History-only OR	Stationary (r, Q) from demand and lead-time history, mean lead time	None	ERP-independent baseline
Croston/SBA OR	(r, Q) with the Syntetos–Boylan-approximated Croston demand rate and its forecast-error spread, mean lead time	None	Intermittent-demand baseline
Empirical-quantile OR	(r, Q) with the reorder point set to the service-level quantile of historical rolling lead-time demand	None	Distribution-free baseline
Simulation-tuned (r, Q)	Reserve factor and order-quantity multiple chosen by simulation on the last 250 working days before the cutoff	None	Trained data-driven baseline
LLM (runs 1–3)	LLM proposal with all CSV fields (Section 5.2)	Both	Main treatment
LLM without ERP	LLM with ERP safety stock and safety lead time set to "not supplied"	None	Ablation: what the LLM does when it cannot anchor on ERP

Three further arms use no ERP input and represent stronger data-driven practice. The *Croston/SBA OR* replaces the mean daily demand by the Croston estimate with the

Syntetos–Boylan bias correction [64, 65] (smoothing constant 0.1), uses the root mean squared one-step forecast error as demand spread and the history-only lead-time moments, which suits intermittent demand [66]. The *empirical-quantile OR* sets the reorder point to the service-level quantile of historical rolling lead-time demand over the rounded mean lead time. The *simulation-tuned* (r, Q) policy selects, from a grid of reserve factors and order-quantity multiples around the history-only policy, the cheapest candidate whose fill rate on the last 250 working days *before the cutoff* meets the service target (or the best-fill candidate if none does). No post-cutoff data are used in any of these arms. We do not implement a deep reinforcement-learning baseline [38, 37]; this is a limitation (Section 7).

5.4. Simulation, capacity replay and metrics

Each pair is simulated independently on the post-cutoff horizon as a single-item, single-echelon continuous-review (r, Q) system [71, 83, 84, 73, 72] with lost sales [85], in the spirit of simulation-based experiments on supply and timing uncertainty [86], a shared opening stock, an empty pipeline and arrivals after the rounded mean lead time in working days (Appendix Appendix A). MOQs are treated as hard order-size constraints, as imposed by suppliers [87], and storage limits as capacity bounds on the replenishment decision [88]. The primary metrics are the mean material-horizon holding-plus-ordering cost, the mean material-level fill rate, the aggregate (volume-weighted) fill rate and the number of stockout days. We report paired differences with 10,000-sample bootstrap 95% intervals.

A static check $SS + Q \leq$ `max_storage_units` does not show that storage limits hold as inventory evolves, because the reorder point includes lead-time demand and inventory on order can exceed the limit on arrival. We therefore record the post-receipt on-hand stock for every policy and day and report the share of pairs and days above the limit (the *unconstrained* simulation), and we *replay* every policy with each order truncated to the storage limit minus

the current inventory position (the *hard-limit replay*), which is our primary evaluation.

5.5. Repeated runs and sensitivities

To examine run-to-run variation, the entire cohort is generated three times (runs 1, 2 and 3) with the same prompt file and the same data. Run 1 serves as the reference for tabulated results unless stated otherwise; runs 2 and 3 provide replicates. Sensitivity analyses vary the working-day reconstruction (majority, any-plant, strict-all, exclusion of Plant C and calendar-day arrivals), the shortage penalty (zero, one and five times unit cost).

6. Results

6.1. Completion and policy characteristics

All four runs (three with the full input and one ablation, Section 6.4) of the original data return a valid, storage-feasible (r, Q) policy for all 365 pairs after at most three attempts per pair. The validity gate rejects non-positive reorder points and order quantities, negative safety stocks and reorder points below the safety stock. The ERP safety stock is zero for 198 of the 365 pairs (188 of the 346 in the cohort below), so for a majority of pairs the SAP-derived arm holds no safety stock at all.

6.2. Headline comparison with ERP-derived parameters

Unless stated otherwise, all results use the hard-limit replay on the 346-pair common-feasible cohort with working-day receipts and no shortage penalty (Section 6.5 gives the unconstrained simulation). Table 3 compares all arms; its last columns give the LLM cost relative to each arm in the three runs (positive: the LLM costs more). In run 1 the LLM policies cost 11.4% less than SAP-derived policies and 30.5% less than the SLT-informed OR (mean 2,928 versus 3,305 and 4,211), at a mean fill 0.18 points lower than SAP-derived

(98.11% versus 98.30%; interval −0.54 to +0.10). The mean cost difference against SAP-derived is −377 (95% interval −651 to −154); the LLM is cheaper in 241 of 346 pairs and, in 226, simultaneously no more expensive and no worse in fill. It incurs more stockout days (910 versus 804).

Relative to SAP-derived parameters the LLM carries about 17% less average inventory (16,256 versus 19,565 units) and about 16% lower holding cost at about the same number of orders per year (17.3 versus 17.6), and it accepts 11–20% more stockout days (runs 3 and 2). The saving is therefore a trade of inventory for service, not a better ordering rhythm, and Section 6.10 shows that once shortages carry a cost this trade no longer favours the LLM.

Table 3: Hard-limit replay, common-feasible cohort ($n = 346$), working-day receipts, zero shortage penalty. Cost is the mean material-horizon holding-plus-ordering cost; average inventory is in units; orders are per year. The last column gives the LLM cost relative to the arm in runs 1/2/3 (positive: the LLM costs more).

Arm	Mean cost	Holding	Ordering	Avg. inv.	Orders	Mean fill (%)	Aggr. fill (%)	Stockout days	LLM vs arm (%)
ERP-derived (SAP)	3,305	2,172	1,132	19,565	17.6	98.30	79.34	804	−11.4 / −11.3 / −11.4
SLT-informed OR	4,211	2,608	1,603	21,614	24.5	98.23	79.15	831	−30.5 / −30.4 / −30.5
ERP-floor OR	4,248	2,633	1,616	21,937	24.7	98.23	79.15	831	−31.1 / −31.0 / −31.1
History-only OR	2,764	1,725	1,039	15,972	16.3	98.14	78.72	874	+5.9 / +6.0 / +5.9
Croston/SBA OR	2,705	1,665	1,040	15,521	16.3	98.30	78.87	803	+8.2 / +8.4 / +8.3
Empirical-quantile OR	2,489	1,543	946	13,331	14.8	98.35	78.87	747	+17.7 / +17.8 / +17.7
Simulation-tuned (r, Q)	3,266	2,473	793	11,914	12.6	96.05	84.84	1,973	−10.4 / −10.3 / −10.3‡
LLM (run 1)	2,928	1,824	1,104	16,256	17.3	98.11	78.87	910	
LLM (run 2)	2,931	1,850	1,081	16,599	17.0	97.89	78.70	968	
LLM (run 3)	2,929	1,841	1,088	16,958	17.0	98.16	78.90	890	
LLM without ERP	3,014	1,902	1,112	16,507	17.3	98.06	78.89	890	−8.8† (vs SAP)

† Other comparisons of the ablation are in Section 6.4. ‡ Different operating point (mean fill 96.05%, aggregate fill 84.8%); the paired interval of the cost difference includes zero, and with priced shortages this policy is the cheapest arm (Section 6.10).

6.3. Information asymmetry: what the LLM adds beyond the ERP inputs

First, the SLT-informed OR and its ERP-floor variant are the most expensive arms (4,211 and 4,248): the ERP safety lead time inflates their effective lead time and buffers. Against them the LLM is 30–31% cheaper in every run, which mostly measures how heavily ERP inputs over-buffer.

Second, the *history-only OR*, which uses no ERP inputs, costs 2,764 at a fill of 98.14%. The LLM costs 5.9–6.0% more in all three runs (run 1: +164, interval +91 to +240), with fill differences of −0.03, −0.25 and +0.01 points whose intervals include zero, and is cheaper

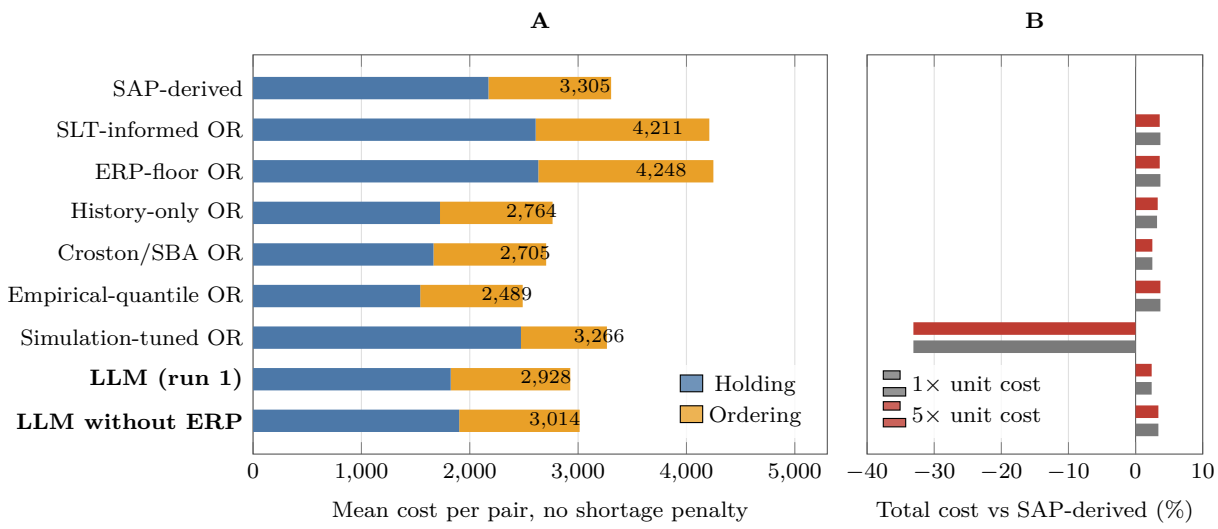


Figure 2: Cost structure and effect of pricing shortages (hard-limit replay, 346 pairs; LLM rows in bold). (A) Mean cost per pair split into holding and ordering cost, with the total at the end of each bar. The LLM and the data-driven ORs carry less inventory than SAP-derived parameters and the ERP-informed ORs. (B) Total cost when lost sales are charged at one and five times the unit cost, relative to SAP-derived parameters (negative: cheaper). Once shortages are priced SAP-derived parameters beat the LLM, and the simulation-tuned policy is cheapest.

in only 74–77 of 346 pairs. The stronger data-driven arms widen the gap: the LLM costs 8.2–8.4% more than the Croston/SBA OR (2,705, fill 98.30%; run 1: +223, interval +97 to +362; cheaper in 91–99 pairs) and 17.7–17.8% more than the empirical-quantile OR (2,489, fill 98.35%; +439, interval +329 to +557; cheaper in 44–49 pairs; mean fill 0.24 points lower). Part of the LLM's advantage over ERP-derived parameters therefore reflects weak ERP benchmarks.

The simulation-tuned (r, Q) policy sits at a different operating point: an aggregate fill of 84.8% and the lowest inventory (11,914 units) but a mean fill of 96.05% and 1,973 stockout days, because it is tuned on the pre-cutoff year. The LLM costs 10.3–10.4% less at zero penalty with a 2.1-point higher mean fill, but the interval includes zero (−2,527 to +857); once shortages are priced the tuned policy is by far the cheapest arm (Section 6.10).

6.4. LLM without ERP planning parameters (ablation)

In this ablation the LLM receives the same inputs as in runs 1–3, but the ERP safety stock and safety lead time are blanked and marked "not supplied"; the system prompt is unchanged. It tests whether the LLM anchors on the ERP planning parameters. All 365 pairs returned a valid, storage-compliant policy, and the policies change only slightly beyond run-to-run variation (median ratios to run 1 of 1.00; the reorder point differs by more than 10% in 54.6% of pairs, against 48.3% and 48.0% between ERP-supplied runs). In the hard-limit replay the cost is 3,014 at a mean fill of 98.06%, against 2,928–2,931 in runs 1–3: 8.8% below SAP-derived parameters (interval −566 to −61), but 9.0% above the history-only OR (interval +134 to +387), 11.4% above the Croston/SBA OR and 21.1% above the empirical-quantile OR. By plant the pattern is unchanged (−10.8% in Plant A, −21.9% in Plant B, +41.6% in Plant C), so the Plant C loss does not stem from reliance on ERP parameters. The ERP planning fields appear to be of limited use to the LLM; the ablation withholds only these two fields, and unit cost, holding rate, ordering cost, MOQ, service target and storage limit remain available, as they are for all reference arms.

6.5. Storage limits along simulated trajectories

All 365 LLM policies satisfy the static rule $SS + Q \leq$ `max_storage_units`, yet in the unconstrained simulation post-receipt on-hand stock exceeds the limit for 183 pairs (52.9%; 178–183 across runs; 10.2% of pair-days) for the LLM, 198 (57.2%) for SAP-derived and 230 (66.5%) for the SLT-informed OR, because inventory on order and lead-time demand add to stock above $SS + Q$. The static check is thus necessary but not sufficient for any method. Under the hard-limit replay no arm exceeds the limit by more than one unit; enforcing it raises the LLM mean cost from 2,850 to 2,928 and SAP-derived from 3,184 to 3,305 without changing the ranking.

6.6. Plants and demand regimes

The average hides a strong plant effect (Figure 3, hard-limit replay, run 1). The LLM costs 14.1% less than SAP-derived in Plant A (cheaper in 179 of 215 pairs) and 22.7% less in Plant B (54 of 65), but 36.2% *more* in Plant C (cheaper in 8 of 66); the Plant C loss is similar in the other full runs (+32.2%, +31.5%) and persists under every working-day reconstruction in which Plant C can be scored (Section 6.10). Against the data-driven baselines the LLM costs more in every plant: 7.2%, 3.6% and 6.7% more than the history-only OR, 13.1%, 2.3% and 6.2% more than the Croston/SBA OR, and 20.8%, 7.7% and 29.5% more than the empirical-quantile OR in Plants A, B and C. The SLT-informed and history-only ORs coincide in Plant C because the ERP safety lead time is zero for all its pairs.

By demand regime, the 33 lumpy materials show the largest reduction against SAP-derived (−25.4% in run 1), but the history-only and Croston/SBA ORs cost about 9–10% less than the LLM there; with 33 materials this is indicative only.

6.7. Why the LLM does not beat code

The prompt prescribes an economic order quantity with a minimum-order-quantity floor and a normal-approximation safety stock (Appendix Appendix A). The History-only OR applies these formulas as code to the same demand and lead-time history, and the SLT-informed and ERP-floor ORs apply them with the ERP inputs the LLM also receives (Section 5). The comparison with code is therefore direct.

The LLM reproduces the formulas.. Pair by pair, the median ratio of the LLM order quantity to the History-only order quantity is 1.00 (medians of 2,810 and 2,828 units) and that of the safety stock is 1.00 (712 and 718 units, against an ERP median of 0). The reorder point is 9% higher than the code's in the median pair, in every segment of Table 4, and the LLM

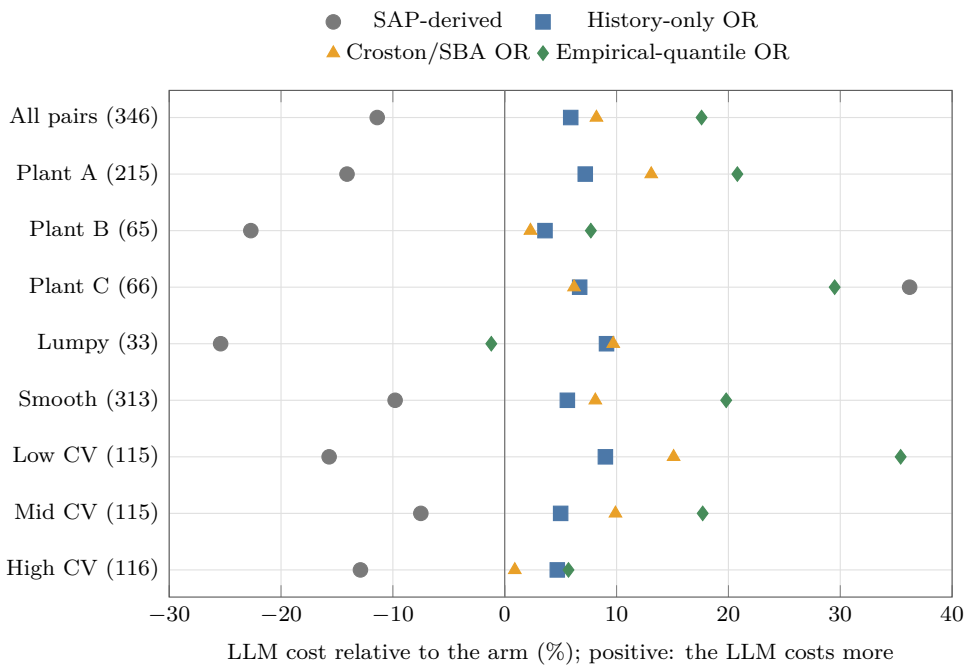


Figure 3: Where the LLM wins and loses (run 1, hard-limit replay). Each marker is the LLM's mean cost relative to one comparator in a segment of pairs (number of pairs in brackets); markers to the right of the vertical line mean the LLM costs more. The LLM is cheaper than SAP-derived parameters in Plants A and B but not in Plant C, and costs more than the three data-driven baselines in almost every segment.

costs 4–9% more than the History-only code in every segment. The difference to code is thus a small, systematic uplift of the reorder point rather than a different policy.

Inputs, not the LLM, decide the ranking against code.. Code that receives the same ERP inputs as the LLM costs 4,211 (SLT-informed OR) and 4,248 (ERP-floor OR) per pair against 2,928 for the LLM in run 1, i.e. 44–45% more than the LLM, because the ERP safety lead times inflate the textbook reorder point. Code that ignores the ERP planning inputs costs 2,764, 5.9% less than the LLM. The LLM resembles the History-only code rather than the ERP-inflated code, which is consistent with the ablation in Section 6.4.

Run-to-run noise.. The cost of a pair varies by 10.5% of its mean between the three runs, and the runs disagree on whether the LLM beats the History-only code for 34.1% of the pairs. The LLM mean is cheaper than the History-only code for 17.1% of the pairs and than SAP-derived parameters for 76.3%. The best of the three runs for each pair, possible only

with hindsight, costs 2,765, equal to the History-only code (2,764); the worst costs 3,146. Repeatability, which code has by construction, is a practical cost of using an LLM here.

Plant C and segments.. Table 4 shows that the LLM advantage over SAP-derived parameters disappears in storage-bound pairs (+4.9% against −14.9% elsewhere) and reverses in Plant C (+33.3%); it does not vary monotonically with demand variability. Plant C is not an LLM-specific failure: the History-only code is also 28% more costly than SAP-derived parameters there, so the ERP parameters are leaner than any formula-based policy in that plant when lost sales are not charged. Plant C also has the least regular demand (median coefficient of variation of daily demand 0.61, lead-time coefficient of variation 0.78, against 0.51 and 0.63 over all pairs).

Table 4: Where the LLM differs from code. Mean cost per pair in the hard-limit replay with zero shortage penalty (LLM: mean of three runs; History-only: textbook code on demand and lead-time history; SAP: ERP-derived), the LLM cost relative to each (in %), and the median ratio of the LLM reorder point to the textbook reorder point. Storage-bound pairs are those for which the textbook safety stock plus order quantity exceeds the storage limit, so that the textbook policy is projected down to it.

Segment	n	LLM	History-only	SAP	vs History-only (%)	vs SAP (%)	ROP ratio
All pairs	346	2,929	2,764	3,305	+6.0	-11.4	1.09
Plant A	215	2,370	2,231	2,782	+6.2	-14.8	1.08
Plant B	65	5,050	4,746	6,360	+6.4	-20.6	1.10
Plant C	66	2,662	2,549	1,997	+4.4	+33.3	1.11
Demand CV, lowest third	115	2,249	2,092	2,707	+7.5	-16.9	1.09
Demand CV, middle third	115	3,872	3,659	4,152	+5.8	-6.8	1.10
Demand CV, highest third	116	2,669	2,544	3,057	+4.9	-12.7	1.08
ERP safety stock zero	188	2,946	2,741	3,215	+7.5	-8.4	1.09
ERP safety stock positive	158	2,909	2,792	3,411	+4.2	-14.7	1.09
Storage-bound	145	1,491	1,408	1,422	+5.9	+4.9	1.10
Not storage-bound	201	3,967	3,743	4,663	+6.0	-14.9	1.08

6.8. Run-to-run stability

Aggregate results are very similar across the three runs (Figure 4): the LLM cost relative to SAP-derived is −11.4%, −11.3% and −11.4%. Individual policies vary, however: between runs 2 and 3 all three values are identical for only 36 of 365 pairs. A single generation per material is a noisy estimate that should be reviewed [14].

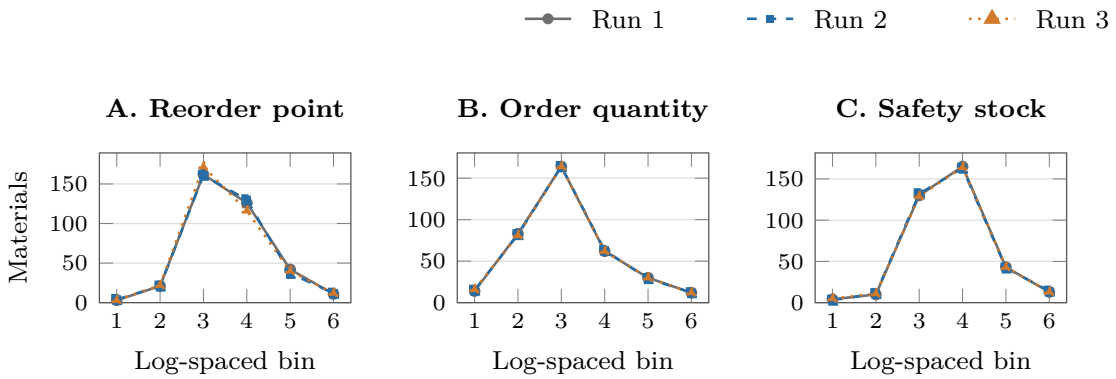


Figure 4: Distribution of the proposed reorder point (A), order quantity (B) and safety stock (C) over the 365 materials in the three runs, on six common log-spaced bins (reorder point 3–683,000; order quantity 63–722,000; safety stock 0–289,000 units). Nearly coinciding lines show reproducible aggregate behaviour.

6.9. Replication on the masked release

The raw data cannot be shared, so we publish a masked release of the panel (see Data and Code Availability): all quantities and prices are multiplied by secret per-pair factors, identifiers are replaced by pseudonyms, and dates and plant structure are unchanged. We repeated the generation three times on the masked panel with the unchanged prompt file (runs m1–m3), and scored the output with the same replay as above. Run m1 produced 364 and runs m2 and m3 all 365 of the 365 policies (one pair failed all three attempts in m1); 340, 342 and 342 policies were scorable, and 5, 4 and 4 were rejected by the validity gate. Table 5 reports the replication.

Table 5: Replication on the masked release: hard-limit replay, working-day receipts, zero shortage penalty. Cost is the mean material-horizon holding-plus-ordering cost in masked currency units; the last rows show the LLM cost relative to each arm (%). Ratios to the original runs use the same pairs.

Quantity	**m1**	**m2**	**m3**
Scored pairs	340	342	342
LLM mean cost	2,892	2,947	2,903
LLM mean fill (%)	98.34	98.27	98.27
LLM vs ERP-derived	−12.1	−10.9	−12.2
LLM vs history-only code	+5.1	+6.4	+5.0
LLM vs Croston/SBA code	+7.4	+8.7	+7.2
LLM vs empirical-quantile code	+17.0	+18.1	+16.5
LLM vs simulation-tuned code	−11.2	−10.1	−11.3
Cost ratio to original run 1	0.991	1.004	0.991

The masked runs reproduce the original findings: the LLM is about 11–12% cheaper

than the ERP-derived parameters, and 5–6%, 7–9% and 16–18% costlier than history-only, Croston/SBA and empirical-quantile code. Mean costs are within 1% of the original runs. By plant, the LLM is 13–16% (PL1) and 21–23% (PL2) cheaper than the ERP-derived parameters but 37–42% costlier in PL3, close to the original figures of −14.1%, −22.7% and +36.2% in Plants A, B and C (plants PL1, PL2 and PL3 correspond to A, B and C). The sign against the simulation-tuned arm differs by plant (about 48–60% costlier in PL1 and PL3, about 52% cheaper in PL2), so its pooled difference depends on the plant mix and is not a stable result.

6.10. Sensitivity analyses

Plant C calendar.. Plant C's working-day flag required reconstruction, so we re-scored all policies under four alternatives (any-plant, strict-all, exclusion of Plant C, calendar-day arrivals). Where Plant C can be scored, the LLM remains 31–35% more expensive than SAP-derived there, so the Plant C result is not an artefact of the repair.

Shortage penalty.. Figure 2B gives mean cost relative to SAP-derived parameters for penalties of one and five times the unit cost. The simulation-tuned policy is then by far the cheapest (about 33% below SAP-derived), because it was selected to meet the service target on the pre-cutoff history; LLM run 1 costs 2.4% more than SAP-derived at both multiples and about as much as the Croston/SBA OR.

Break-even shortage price.. Table 6 re-scores the same replays for a lost-sales penalty between zero and one unit cost per unit. The LLM is cheaper than SAP-derived parameters only while the penalty stays below 0.9% of the unit cost, and cheaper than the History-only and empirical-quantile code only above 1.5% and 2.4%, by about 1% or less. It is cheaper than the SLT-informed and ERP-floor ORs at every penalty and cheaper than the simulation-tuned

policy only below 0.06%. From a penalty of 1% of the unit cost, the SAP-derived, History-only, Croston/SBA, empirical-quantile and LLM policies lie within 4% of each other, while the simulation-tuned policy is 28–33% cheaper than SAP-derived parameters. The ranking between the LLM and the formulas is thus decided by a shortage price that the primary basis sets to zero, which is why it is reported as a sensitivity and not as a headline result.

Table 6: Break-even shortage price. Mean total cost per pair (holding, ordering and shortage cost) in the hard-limit replay when each lost unit is charged at the given multiple of the unit cost, and the break-even multiple at which the LLM and the arm have equal mean cost (common-feasible cohort, $n = 346$).

Arm	0	0.01	0.1	1	Break-even multiple of unit cost
SAP-derived	3,305	19,613	166,388	1,634,136	0.0089; LLM cheaper below
History-only OR	2,764	19,609	171,207	1,687,191	0.0145; LLM cheaper above
Croston/SBA OR	2,705	19,427	169,927	1,674,925	none; LLM costlier on the grid
Empirical-quantile OR	2,489	19,402	171,618	1,693,784	0.0242; LLM cheaper above
Simulation-tuned	3,266	14,174	112,344	1,094,043	0.0006; LLM cheaper below
LLM (mean of three runs)	2,929	19,660	170,238	1,676,014	–

7. Discussion, Limitations and Research Perspectives

What the evidence shows.. A prompt-configured LLM generates feasible, reviewable (r, Q) policies for every material; in simulation they cost less than ERP-derived parameters but more than data-driven OR policies. The mechanism analysis (Section 6.7) explains why: the LLM applies the textbook recipe it is given, with a small reorder-point uplift and run-to-run noise, so it cannot beat the same formulas run as code on the same inputs. InventOR is therefore a reviewed candidate generator, not an optimizer. For an organization that only needs parameters, running those formulas, or a tuned policy once shortages are priced, is the cheaper and exactly repeatable choice; the LLM is justified where a written rationale, an escalation flag and the absence of coding effort matter more than a few percent of holding and ordering cost. Whether that helps planners is a hypothesis the simulation cannot test.

Implications for practice.. (1) Benchmark against data-driven code, not only against ERP values: ERP-derived parameters are weak benchmarks (stale or zero safety stocks, inflated

safety lead times), and comparing only with them overstates the benefit. (2) Price shortages before reading cost comparisons, since without a penalty the ranking rewards under-stocking (Section 6.10). (3) Verify storage along the trajectory: the static rule $SS + Q \leq$ limit was breached in simulation for 53–66% of pairs under each arm checked (Section 6.5); enforce a cap in the order-release logic or treat the limit as a design requirement. (4) Expect plant-level differences (Section 6.6) and define in advance when proposals are trusted, escalated or rejected, to limit complacency [11]. (5) Treat the system prompt as a controlled file, log every call and repeat runs before changing it (Section 6.8). (6) Generation took several minutes per material against milliseconds for OR baselines; periodic regeneration could support adaptable planning [89]. Informal feedback from the responsible planner was positive, but it is not a validation.

Limitations.. (i) The data come from one manufacturer and three plants. (ii) The model identifier and inference settings are operator-declared. (iii) No planner outcomes were measured. Informal feedback from the responsible planner was positive; it is anecdotal and not evidence of benefit, and the benefits of reviewability, auditability and planner support are design intentions that a field study must test. (iv) The simulation uses deterministic arrival at the rounded mean lead time, ignores observed receipts, starts with an empty pipeline and does not simulate lead-time variability, which may bias comparisons that depend on safety lead time. (v) The primary results use a zero shortage penalty; with a positive penalty the ranking changes. Costs are dominated by a few high-demand materials and the 346 pairs are not independent, so the bootstrap intervals understate uncertainty. (vi) The (r, Q) family is single-item and single-echelon, and individual policies vary across runs. (vii) We included no deep reinforcement-learning or neural forecasting baseline, and the ERP-derived arm (ERP safety stock with our own order quantity) is not an observed record of practice. (viii) Con-

fidential data prevent replication; we release code, prompts and aggregate results, and an open-dataset replication with open-weight models is the most direct next step.

Research perspectives.. (1) *Ensembling and hybrids:* aggregating several generations may reduce individual-policy noise [34], and combining the best data-driven policy with LLM review [90] could capture both the OR cost level and the LLM's flags. (2) *Comparison with trained methods:* benchmarks against reinforcement-learning policies [43, 37, 38] and richer policy families such as (s, S) [67, 68] on identical data would clarify the trade-off between setup effort and policy quality. (3) *Planner-in-the-loop field trials:* prospective studies should compare planners with and without the aid on matched materials, measuring planner time, override rates and realized service and inventory, and examine adoption drivers [22, 21, 6].

8. Conclusion

We evaluated whether a prompt-configured LLM can set replenishment parameters, a recurring production-planning decision, for 365 plant-material pairs of an industrial manufacturer. Without training, the LLM generated feasible, justified (r, Q) policies for every material, and a deterministic gate, a hard-limit simulation and explicit review rules made them checkable. The proposals cost about 11% less than ERP-derived parameters at slightly lower service, but the LLM costs about 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile policies; when lost sales are priced at one unit cost or more, ERP-derived parameters are the cheapest of the non-tuned arms and a simulation-tuned policy is cheapest overall. Storage limits that hold in a static check are exceeded along simulated trajectories for every method, and the LLM loses to ERP at one plant. The LLM reproduces the textbook formulas it is given, so the same formulas run as code are cheaper and exactly repeatable, and the ranking against code is decided by a shortage price that

the primary basis sets to zero. The evidence supports InventOR as a checkable candidate generator inside a reviewed process, not as an optimizer that beats data-driven baselines; field evidence on planner behaviour and realized outcomes is the central open step.

Appendix A. Formal Definitions

Reference calculation.. These are the standard textbook calculations [73, 83]. For pair i, let $\bar{d}_i$ and $\sigma_{D,i}$ be the mean and standard deviation of pre-cutoff working-day demand, L_i and $\sigma_{L,i}$ the mean and standard deviation of the lead time in working days, K_i the ordering cost, h_i the annual holding cost per unit (unit cost times holding rate), z_i the standard normal quantile of the service target, and 250 the working days per year. The order quantity is $Q_i = \max\{\mathrm{MOQ}_i, \lceil\sqrt{2K_i \cdot 250\bar{d}_i/h_i}\rceil\}$. The lead-time-demand standard deviation is $\sigma_{LTD,i} = \sqrt{L_i\sigma_{D,i}^2 + \bar{d}_i^2\sigma_{L,i}^2}$, the safety stock $SS_i = \lceil z_i\sigma_{LTD,i}\rceil$ and the reorder point $R_i = \lceil \bar{d}_i L_i^{\mathrm{eff}} + SS_i\rceil$, where L_i^{eff} is the mean lead time for the history-only OR and the mean lead time plus the ERP safety lead time for the SLT-informed OR and the ERP-floor OR. The SAP-derived arm uses the ERP safety stock and the same Q_i. All policies are projected to $SS_i + Q_i \le \texttt{max_storage_units}_i$ when $\mathrm{MOQ}_i + SS_i$ fits.

Simulation.. On each post-cutoff working day, demand is served from on-hand stock, unmet demand is lost, and an order of size Q_i is placed when the inventory position (on-hand plus on order) is at or below R_i and arrives after the rounded mean lead time in subsequent working observations. Observed receipts, corrections and scrap are ignored. The total cost is $TC_i = h_i\bar{I}_i T/250 + K_i N_i$, where $\bar{I}_i$ is the mean on-hand stock over the T simulated working days and N_i the number of orders. Fill rate is served divided by demanded units. In the hard-limit replay an order is truncated to $\texttt{max_storage_units}_i$ minus the inventory position, and no order is placed if the truncated quantity is zero.

Author Contributions

Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Validation, Visualization, Writing – original draft, Writing – review and editing: Aris Dressino.

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Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

The raw operational data and material-level artifacts cannot be publicly released because they contain commercially sensitive material, supplier and planning information. The company approved the release of a *masked panel* of the same 233,078 records in which identifiers are pseudonymized and quantities and prices are multiplied by secret per-pair factors (the factors are not released). The panel, the frozen outputs of the original runs rescaled to it, the three masked re-runs (Section 6.9), the analysis code, the aggregate evaluation results, manuscript sources and prompt files are available at `github.com/DressPD/inventor_public`. A versioned Zenodo archive of the release is available at `https://doi.org/10.5281/zenodo.22262051`. Supplementary File S1 contains the system prompt used for all runs; the prompt file and, for each call of the masked runs,

a hash of the input, the UTC time and a hashed session identifier are in the release. The runner code hash was re-stamped for 38 records of run m1 after a change that only affected authentication handling (stop on an expired token); prompt, input, model settings and call logic are unchanged, and the original hash is kept in each amended record. The model identifier and inference settings are not exposed by the platform and are operator-declared (Anthropic Claude Sonnet 4.6).

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Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the author used OpenAI GPT-5.6 and Anthropic Claude (Sonnet 5.5), accessed through the OpenCode coding assistant, in order to edit language, restructure text, check consistency, verify citations against Crossref metadata, proof the LaTeX source, and write and run the analysis scripts that produce the reported evidence. These assistants are distinct from the Claude Sonnet 4.6 deployment evaluated in the study. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article. The figures are data visualizations drawn by scripts directly from the reported evidence; no generative AI created or altered any image.

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